















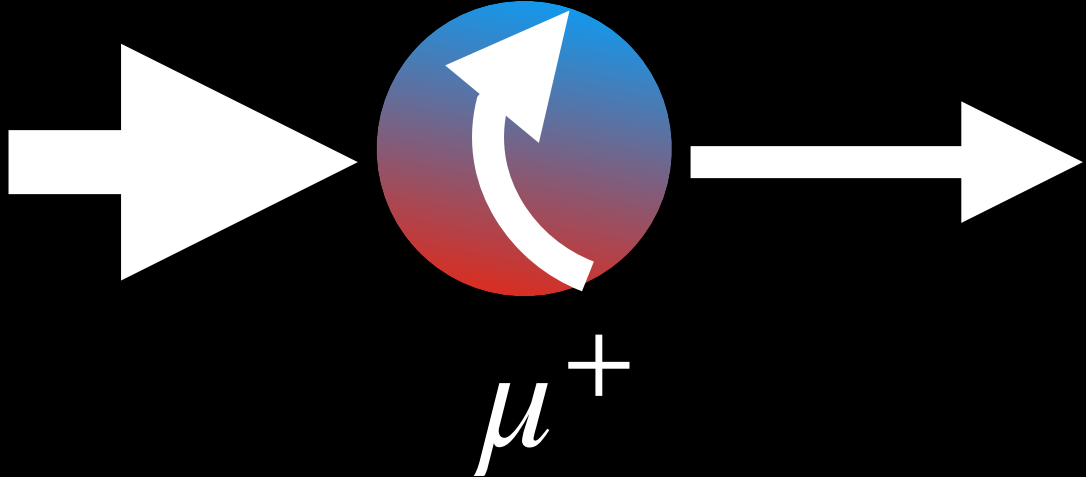




# The experiments were repeated!

At the Christmas party in 1956 Lederman learned about Wu's results.

Leon Lederman, Richard Garwin and Marcel Weinrich performed an experiment at particle accelerator

If parity is violated, then  $\mu^+$  in  $\pi^+ \rightarrow \mu^+ + \nu$  decay is polarized   $\mu^+$

The diagram shows a large white arrow pointing right, followed by a sphere with a blue-to-red gradient. Inside the sphere is a white curved arrow indicating a counter-clockwise rotation when viewed from the right. A smaller white arrow points right from the sphere. Below the sphere is the label  $\mu^+$ .

$\mu^+$  polarization can be measured in  $\mu^+ \rightarrow e^+ + 2\nu$  decay as  $e^+$  asymmetry

Less than in a week in early 1957 they observed parity violation in  $\mu^+ \rightarrow e^+ + 2\nu$

**Parity violation was firmly  
established**